AI651 Assignment 2 Report

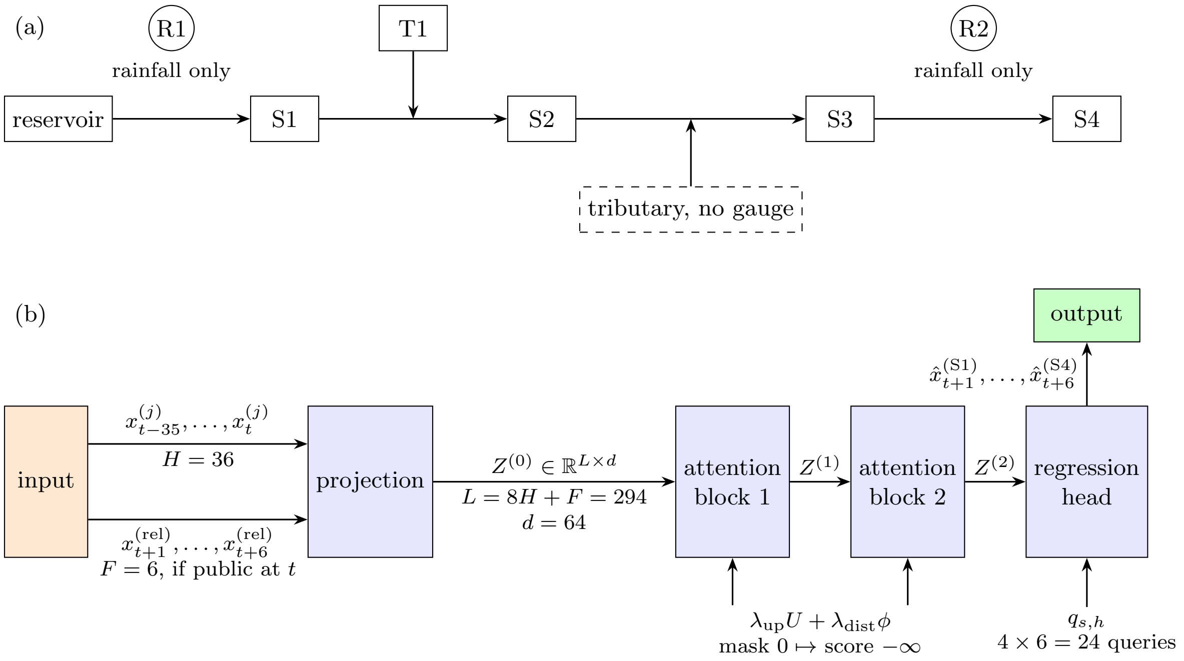
Muhammad Tayyab Tahir Qureshi
Roll number 25280024
Lahore University of Management Sciences

Fall 2026

Name: Muhammad Tayyab Tahir Qureshi
Roll number: 25280024
GitHub repository (public): `https://github.com/ttqureshi/DL4STG-PA2`

1 Question 1

1.1 Architecture diagram


(a)
R1
rainfall only
T1
R2
rainfall only
reservoir
S1
S2
S3
S4
tributary, no gauge
(b)
output
$\hat{x}^{(\mathrm{S1})}_{t+1}, \ldots, \hat{x}^{(\mathrm{S4})}_{t+6}$
input
$x^{(j)}_{t-35}, \ldots, x^{(j)}_{t}$
$H = 36$
$x^{(\mathrm{rel})}_{t+1}, \ldots, x^{(\mathrm{rel})}_{t+6}$
$F = 6$, if public at t
projection
$Z^{(0)} \in \mathbb{R}^{L \times d}$
$L = 8H + F = 294$
$d = 64$
attention block 1
$Z^{(1)}$
attention block 2
$Z^{(2)}$
regression head
$\lambda_{\mathrm{up}} U + \lambda_{\mathrm{dist}} \phi$
mask 0 $\mapsto$ score $-\infty$
$q_{s,h}$
$4 \times 6 = 24$ queries

(a) River, not to scale; water flows left to right. R1 and R2 are rain gauges only. (b) Forecast map at issuance t on the 10-minute clock. The eight series j and an announced release enter one projection, two attention blocks mix the L tokens, and the regression head issues $\hat{x}^{(s)}_{t+h}$ for $s \in \{\mathrm{S1}, \mathrm{S2}, \mathrm{S3}, \mathrm{S4}\}$ and $h = 1, \ldots, 6$. Future rain and future levels are not inputs.

The eight series are projected and concatenated, and $\hat{Y}$ is the six future levels at S1–S4. $H = 36$, $F = 6$, and $d = 64$ are choices in assumptions A1 and A2.

1.2 Assumptions

The river layout, the recorded series, and the lack of a travel time are given; A1–A16 are not. A2, A10, A14, and A15 are design choices. The others are assumptions about the river or the record.

- **A1.** History is $H = 36$ bins, six hours, ending at issuance. That window has to cover the travel time between neighbouring stations. A longer travel time is seen only through the level it has already produced.
- **A2.** The output is six levels at each of S1–S4, at +10 through +60 minutes, so $\hat{Y}$ has shape $(B, 4, 6)$.
- **A3.** The archive covers at least one year, including wet and dry periods. Inference still uses only the 36-bin window.
- **A4.** On a normal issuance the bin that closes at t has arrived. If it has not, it stays masked and the forecast still goes out.
- **A5.** Gauge errors are small beside a rise that matters. There is no separate denoiser.
- **A6.** Each rain reading goes in the 10-minute bin that contains its timestamp. An empty bin repeats the last reading and is marked carried, and a carried bin gets no attention weight, so the repeated value is not read as new rain.
- **A7.** Releases are piecewise constant between effective times, and the latest already-public announcement is used.
- **A8.** S1–S4 and T1 share 10-minute boundaries, so one lag index is the same time at every station.
- **A9.** Water moves downstream, and travel time is not distance divided by a constant speed.
- **A10.** The target is level, and each series is standardized by its training mean and standard deviation.
- **A11.** R1 and R2 are not assigned to reaches; both are tokens, with gauge-to-station distance as a feature.
- **A12.** The unmonitored tributary has no gauge, and the $\mathrm{S3}-\mathrm{S2}$ residual is an extra history channel, zero on the other series.
- **A13.** Future rainfall and future levels are not inputs, and a future release is written only when it was announced by issuance.
- **A14.** Training uses the same clock: inputs at t obey A13, and the target is the level at $t + \Delta, \ldots, t + 6\Delta$.
- **A15.** The loss is mean squared error of the standardized level over stations, leads, and the batch, with missing targets left out.
- **A16.** Along-river distance is used among the reservoir, S1–S4, and T1, and gauge-to-station distance is straight-line. Neither distance is converted into a travel time.

1.3 Forecasting task and data specification

An issuance runs when a 10-minute bin closes at time t. The history is the $H = 36$ bins from $t - 35\Delta$ to t, and the outputs are the six levels at S1–S4 for leads $h = 1, \ldots, 6$. T1 is an input, and the batch size in operation is 1.

Inputs and the forecast at issuance t. Shapes are after alignment and before the width-d projection. Masks are in $\{0, 1\}$.

Name	Shape	Axes	Time	At issuance
Level history	$(B, 5, H)$	batch; S1, S2, S3, S4, T1; history	$t - (H-1)\Delta \ldots t$	Known
Level mask	$(B, 5, H)$	same axes	same bins	Known
S3−S2 residual	$(B, 5, H)$	same; nonzero only on S3	same; 0 if either missing	Constructed
Rain, aligned	$(B, 2, H)$	batch; R1, R2; history	level-clock bins	Past only
Rain mask	$(B, 2, H)$	same axes	1 if a real reading	Known
Actual release	$(B, 1, H)$	batch; release; history	bins ending at t	Known
Schedule on history	$(B, 1, H)$	same axes	already public	Known
Future release	$(B, 1, F)$	batch; release; leads	$t + \Delta \ldots t + 6\Delta$	If announced by t
Future-release mask	$(B, 1, F)$	same axes	1 if public	Known
Upstream U, distances	$(8, 8)$	series by series	static	Known
Forecast $\hat{Y}$	$(B, 4, 6)$	batch; S1–S4; six leads	$t + \Delta \ldots t + 6\Delta$	Target, never an input

1.4 Model computations and design justification

$\Delta = 10$ minutes. Lag $\ell = 0, \ldots, H-1$ means time $t - \ell\Delta$. Lead $h = 1, \ldots, 6$ means time $t + h\Delta$. $U_{j,i} = 1$ when series j is upstream of series i. Into S1: the release only. Into S2: the release, S1, and T1. Into S3: those plus S2. Into S4: those plus S3. A station is not upstream of itself. $\phi(d) = -\log(1 + d/d_0)$, with d_0 the median static distance. In that formula d is a distance; elsewhere $d = 64$ is the token width. L is the token count, not the lookback. λ_{up} and λ_{dist} are learned scalars. λ_{up} biases the score rather than masking it, so a downstream token can still be used if that scalar is small.

A level token carries the standardized level, the mask, and the residual channel. Rain carries standardized rain and its mask. A history release carries actual, schedule, and the actual mask. A future release carries the announced value and the known mask. Each group has its own matrix,

$$e = W_{\text{group}} x + a_i + b_{\text{time}},$$

and the token list is $Z^{(0)} \in \mathbb{R}^{L \times d}$. For block $n = 1, 2$,

$$\text{score}(p, q) = \frac{Q_p \cdot K_q}{\sqrt{d}} + \lambda_{\text{up}} U_{\text{series}(q),\text{series}(p)} + \lambda_{\text{dist}} \phi(d_{q,p}).$$

A key whose mask is 0 has score $-\infty$. The block is residual attention plus a residual feed-forward layer of width $4d$. The query for lead h at station s is $q_{s,h} = W_q[z_{s,0}; e_h; \tau_s]$, where τ_s is how many bins since s last reported. The loss, with $m_{s,h}(t) = 1$ when the target was recorded, is the mean over the batch of

$$\frac{\sum_{s,h} m_{s,h} (\hat{y}^{\text{std}}_{s,h} - y^{\text{std}}_{s,h})^2}{\sum_{s,h} m_{s,h}}.$$

One attention block connects every remaining token, so S4 can see the reservoir in one step, and a second block reuses a mix the first has formed. Attention is used instead of a recurrent stack because that path has to exist inside one issuance. With $L = 294$ and $d = 64$, the projections, the scores, and the width-$4d$ feed-forward layer are about 2.4×10^7 multiply-adds per block. The release is the only upstream series into S1, and T1 enters at S2. Rain and the S3−S2 residual reach a station as tokens in the same attention, and they are not rows of U. Both rain gauges are therefore visible at every station, and the distance feature is what tells them apart. What is fixed is the series list, U, and the rule that an announcement enters only if it is already public. What is learned is every matrix, both blocks, the readout, and the two scalars λ_{up} and λ_{dist}.

1.5 Operation and exceptional conditions

Missing observations. A missing value is set to 0 after standardization and its mask to 0, so that token gets score $-\infty$.

Delayed observations. A bin that has not arrived by t is missing, and the readout sees τ_s, the bins since that station last reported. The forecast is issued on the 10-minute mark and does not wait.

Announced future releases. The latest announcement with announcement time at or before t and effective time in the lead bin has known mask 1, and if none exists the mask is 0. A published flat schedule has mask 1, so "no change" is different from "no announcement."

Scheduled versus actual releases. History bins carry the recorded actual and the announced schedule, and the future channel is only the schedule. The actual release after t is not an input. The next issuance sees it in the history.

Each issuance is one forward pass of two attention blocks over 294 tokens of width 64 and 24 queries, which fits between the ten-minute bins.

1.6 Uncertainty, limitations, and additional information

The readout is a point forecast. The loss is mean squared error, so there is no variance and no quantile. An operator should read a large rise as the warning, and should not read the number as a confidence interval.

The unmonitored tributary can raise S3 and S4 with no trace in S2, T1, the release, or R1 and R2 until that water reaches S3. I use the S3−S2 residual and S3's history, and S4 reads those tokens in one attention block. That residual is not a measured inflow: the gauges have different datums, and the same-time difference ignores the travel lag, so the block learns a mapping from the channel. Inflow that has not reached S3 by t is still unseen.

A changing travel time can put a peak in the wrong 10-minute bin. The model attends over every lag in the $H = 36$ window, with a separate query for each of the six leads, and uses distance as a feature. A travel time longer than H, or a speed change during the future hour, is still unseen.

A gauge on the unmonitored tributary would be a ninth series and would show that inflow before it reaches S3. Travel times at a low stage and a high stage would check whether $H = 36$ cuts off ordinary waves, and a catchment plus an error bar would fix which gauge belongs to which reach and how the loss weights the instruments. S1–S4 are still forecast without those extras.

2 Question 2

Write-up frame f is tensor index $f - 1$. Every learned-model entry is the mean of seeds 0 and 1, with the two seeds in parentheses. Blank and Copy-last are one number each. Test numbers use videos 9000–9999, and every timed run is a Kaggle Tesla T4.

2.1 Table 1

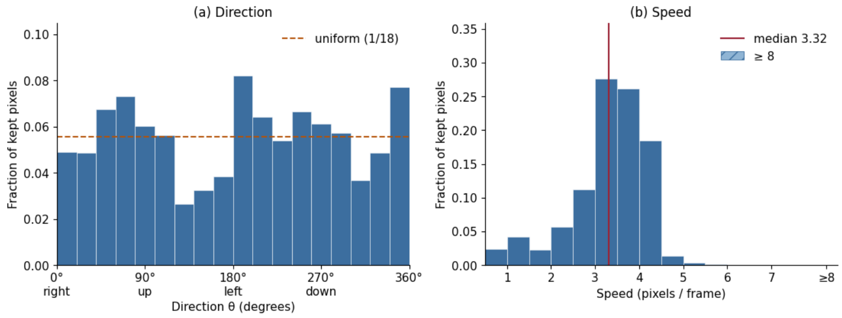
The same mirror, rotation, and window functions are used in the experiments. Seed 0 gives $x = \text{torch.rand}(20, 1, 64, 64)$. The marker p is zero except $p[7, 0, 10, 20] = 1$.

Table 1: Geometry and window checks. Maximum absolute difference is 0 and `torch.equal` is true for (a)–(c).

Check	Expected	Yours
(a) $\max \lvert M(M(x)) - x \rvert$	0	0, equal
(b) $\max \lvert R^4(x) - x \rvert$	0	0, equal
(c) $\max \lvert MRM(x) - R^{-1}(x) \rvert$	0	0, equal
(d) bright pixel of $M(p)$	$(7, 10, 43)$	$(7, 10, 43)$
(e) bright pixel of $R(p)$	$(7, 43, 10)$	$(7, 43, 10)$
(f) 1×, $t = 1$	inputs 1–5, targets 6–10	inputs 1–5, targets 6–10
(g) 2×	1, 3, 5, 7, 9 / 11, 13, 15, 17, 19	same
(h) 3×	1, 4, 7, 10, 13 / 16, 19	same

2.2 Figure 1 and Q2.1–Q2.2

Figure 1. Optical flow on the first 50 MovingMNIST training videos



Kept pixels have intensity > 0.2 in both frames and flow magnitude > 0.5. Median speed = 3.32 pixels/frame. Pixels kept = 103,050. Panel (a): 18 bins of 20°. Dashed line is 1/18. Panel (b): 0.5-wide bins; hatched bar is the overflow bin ≥ 8.

Figure 1: Farneback optical flow on the first 50 training videos, 950 frame pairs. Pixels kept have intensity above 0.2 in both frames and flow magnitude above 0.5. Median speed is 3.32 pixels/frame. Pixels kept: 103,050. Panel (a): 18 bins of 20°, dashed line at 1/18. Panel (b): 0.5-wide bins; the hatched bar is the overflow at or above 8. Every direction bin is occupied, and most of the speed mass sits between 2.5 and 4.5 pixels/frame.

Q2.1. Yes: the mirrored and rotated motion is already familiar, and the new part is the digit shape. Every 20° bin is occupied. The tallest is $[180°, 200°)$ at 0.0820, then $[340°, 360°)$ at 0.0772 and $[60°, 80°)$ at 0.0731, and the shortest is $[120°, 140°)$ at 0.0263. A mirror moves 13.2% of the 103,050 kept pixels into a different bin, and a 90° rotation moves 20.0%. Those pixels still land in a bin the sample already has, so the percentages measure how uneven the histogram is, not a new direction. The median speed is 3.32 pixels/frame, with interquartile range 3.00 to 3.89, so 2× and 3× are 6.64 and 9.95; 0.835 of kept pixels lie between 2.5 and 4.5, the 99th percentile is 4.67, and 0.00465 exceed 5. Where two digits overlap, the flow blends their velocities, and a blend is not faster than the faster digit, so overlap cannot create the missing fast pixels or empty a direction bin.

Q2.2. Mirror changes the glyph and sends θ to $180^\circ - \theta$, and the speed is unchanged. Rotation changes the glyph further and adds 90° to every flow angle, and the speed is unchanged. Speed-up keeps the original frames and multiplies the step between observed frames by two or three. The three tests are useful because each changes one factor: mirror and rotation change appearance and keep the motion, and speed-up changes the motion and keeps the appearance.

2.3 Model A, Table 2, Figure 2, Q2.3–Q2.4

Model A has 480,784 trainable parameters: layer 1 has 184,576, layer 2 has 295,168, and the head has 1,040. On the Tesla T4 a training epoch took 11.3 seconds, and inference on the 1,000 test clips took 0.461 seconds.

Table 2: Test error at 1×, $t = 1$. Model A is the mean (seed 0, seed 1).

	Model A	Blank	Copy-last
MSE_1	0.0169 (0.0174, 0.0164)	0.0426	0.0492
MSE_2	0.0203 (0.0209, 0.0198)	0.0426	0.0604
MSE_3	0.0226 (0.0231, 0.0221)	0.0426	0.0669
MSE_4	0.0244 (0.0248, 0.0240)	0.0425	0.0708
MSE_5	0.0261 (0.0264, 0.0257)	0.0425	0.0732
$E(5)$	0.0221 (0.0225, 0.0216)	0.0426	0.0641
Parameters	480,784	0	0

Figure 2. Model A forecasts on three test videos

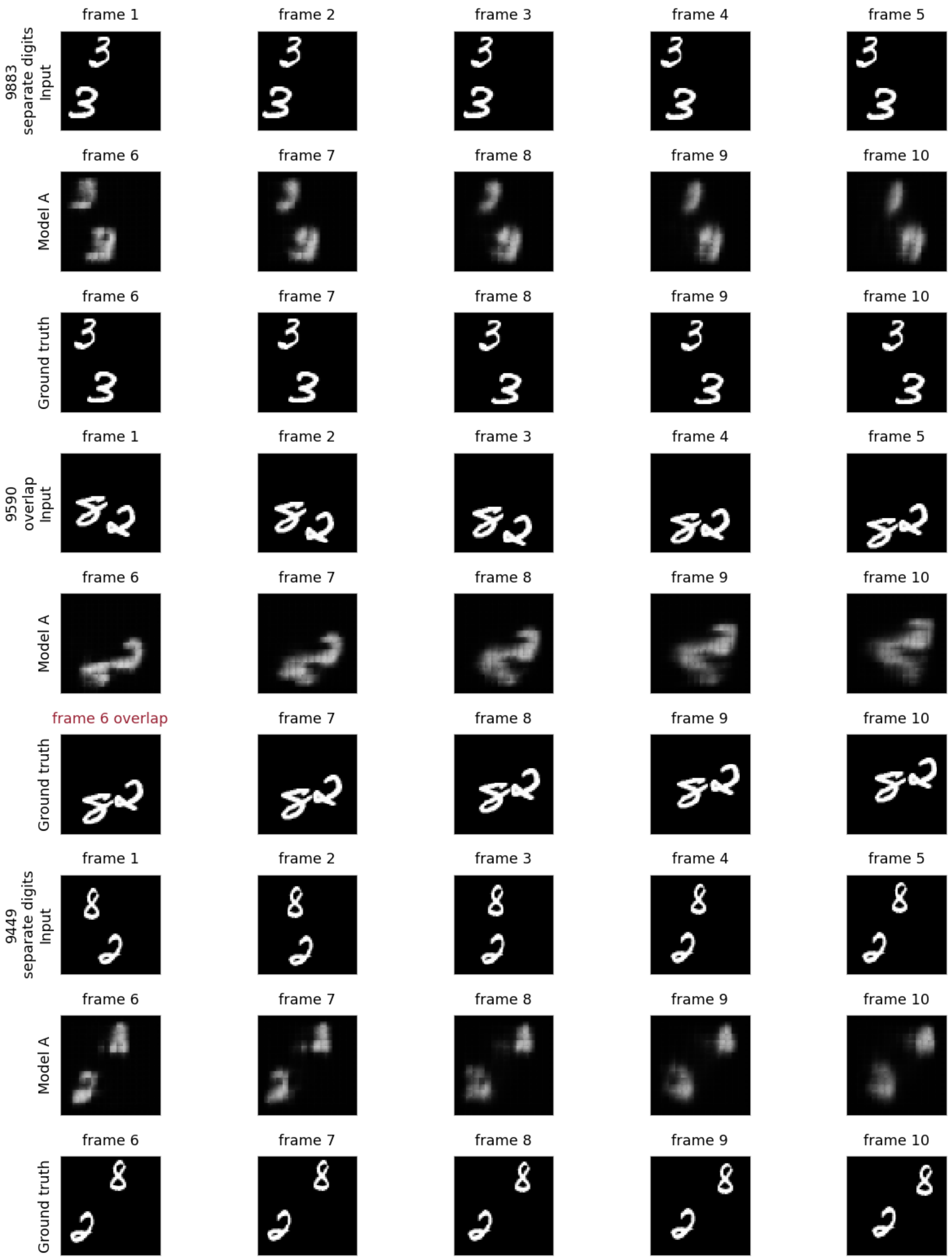


Figure 2. Model A, seed 0, on MovingMNIST test videos 9883, 9590, and 9449 (dataset indices). Each block shows input frames 1–5, the five predicted frames, and ground-truth frames 6–10. Index 9590 has overlapping digits in ground-truth frame 6.

Figure 2: Model A, seed 0, on test videos 9883, 9590, and 9449. Each block shows input frames 1–5, the five predicted frames, and ground-truth frames 6–10. The three videos were chosen from the true frames: 9590 is the overlap case in frames 6–10, and 9883 and 9449 are the two clips whose digits stay furthest apart. The forecast follows separate digits and smears the crossing.

Q2.3. Model A's one-step reach is 8 original pixels. `pixel_unshuffle(4)` leaves a 16×16 grid whose neighbouring cells are 4 original pixels apart, centre to centre. The fold, the 1×1 head, and `pixel_shuffle` do not move information to another cell. Each 3×3 kernel moves one folded cell. Inside one step the new input passes through layer 1 and then through layer 2, which reads layer 1's new hidden state. That is two cells, $2 \times 4 = 8$ original pixels. The previous hidden state of layer 1 travels the same two kernels. The previous hidden state of layer 2 travels only one, which is 4 original pixels. The longest path is $u_\tau \to$ layer-1 $3 \times 3 \to h_\tau^{(1)} \to$ layer-2 $3 \times 3 \to \hat{x}_{\tau+1}$.

Q2.4. Model A's $E(5)$ of 0.0221 is below Blank (0.0426) and Copy-last (0.0641) at every horizon. The seed-mean error grows from 0.0169 at horizon 1 to 0.0261 at horizon 5, and horizon 5 is still below Blank, because each prediction is fed back as the next input and the error carries forward. Copy-last leaves ink where the digit has left. On 9883 and 9449 the forecast is a soft mark near each separate digit; on 9590 the digits cross and the seed-0 forecast is still one smear at frames 9 and 10, where the truth has separated. The smear is the conditional mean of the futures the model cannot tell apart.

2.4 Q2.5

This answer was committed as `10cb5f4` at 14:35 on 27 October 2026, before the mirror, rotation, and speed tests. It is unchanged.

Model A will not pass the mirror test. Its one-step reach is 8 original pixels, which is a statement about how far a 3×3 convolution can move information, not about reflection. The filters are ordinary convolutions, with no mirrored copy of the weights. Figure 1(a) is not symmetric under a horizontal flip: reflecting the measured flow sends θ to $180^\circ - \theta$ and moves 13.2% of the 103,050 kept pixels into a different 20° bin. A mirrored digit is also a different glyph. Both the direction balance and the appearance sit outside the 1× training sample, so $f(M(X))$ should disagree with $M(f(X))$ and with $M(Y)$.

Model A will not pass the rotation test, and rotation should be worse than mirroring. A 90° anticlockwise turn changes the glyph further than a left-right flip, and on these same pixels it reassigns 20.0% of the kept flow into a different bin. Nothing in the two-layer ConvLSTM ties a filter to its quarter-turn.

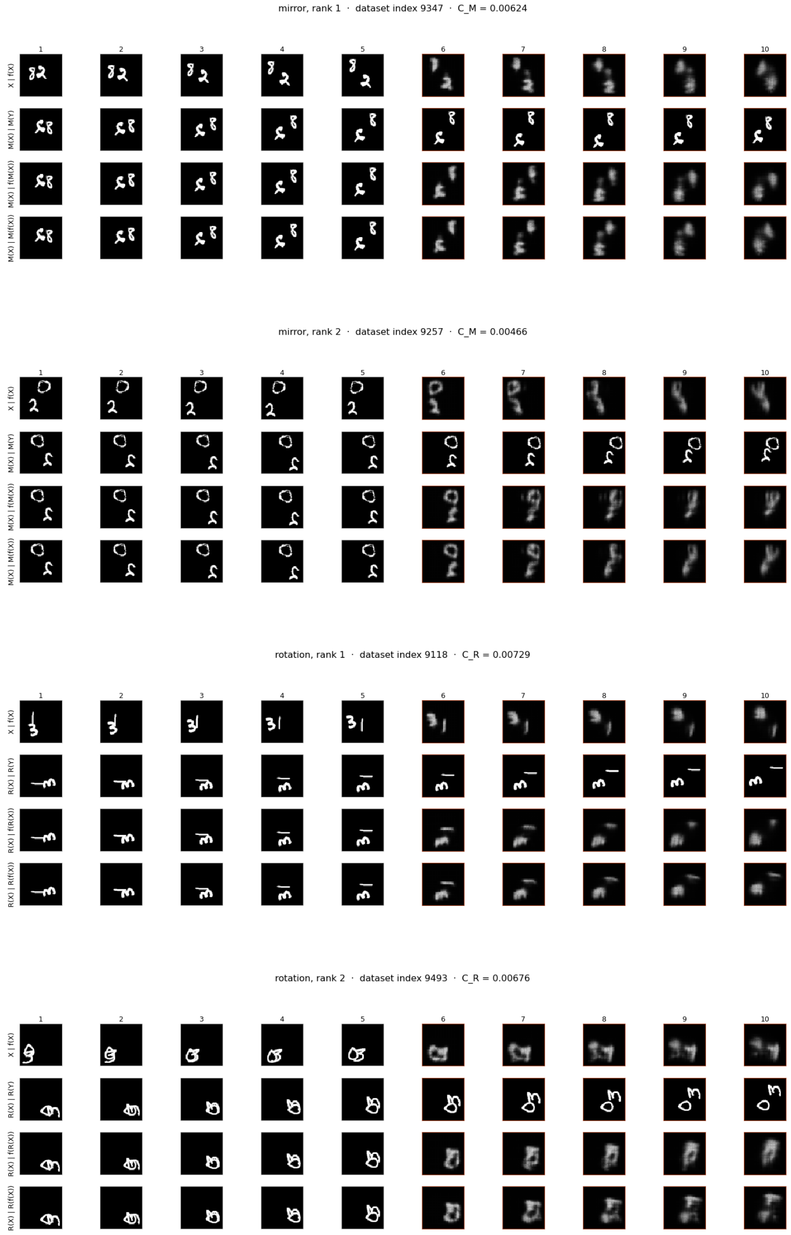
It should break at 3×, and 2× should already be strained for the faster digits. The median consecutive-frame speed is 3.32 pixels per frame, inside the 8-pixel reach. Twice that median is 6.64, still under 8 but close. The 99th percentile at 1× is 4.67, which becomes about 9.3 at 2× and is already past the reach. Three times the median is 9.95, beyond 8.

2.5 Tables 3–4 and Figures 3–5

Table 3: Consistency and transformed error at 1×, $t = 1$. Blank and Copy-last have $C_M = C_R = 0$.

	Model A	Blank	Copy-last
$E(5)$	0.0221 (0.0225, 0.0216)	0.0426	0.0641
C_M	0.00183 (0.00168, 0.00198)	0	0
E_M	0.0227 (0.0232, 0.0223)	0.0426	0.0641
C_R	0.00308 (0.00315, 0.00302)	0	0
E_R	0.0238 (0.0242, 0.0234)	0.0426	0.0641

Figure 3. Seed-0 Model A, largest per-video consistency gaps



mirror, rank 1 · dataset index 9347 · C_M = 0.00624

mirror, rank 2 · dataset index 9257 · C_M = 0.00466

rotation, rank 1 · dataset index 9118 · C_R = 0.00729

rotation, rank 2 · dataset index 9493 · C_R = 0.00676

Figure 3. Seed-0 Model A at 1x, t=1. The two test videos with the largest per-video C_M are dataset indices 9347 (C_M=0.00624) and 9257 (C_M=0.00466). The two with the largest per-video C_R are 9118 (C_R=0.00729) and 9493 (C_R=0.00676). Each panel is four rows by ten frames: columns 1-5 are the input side and columns 6-10 are the future side. Row 1 is X | f(X), row 2 is T(X) | T(Y), row 3 is T(X) | f(T(X)), row 4 is T(X) | T(f(X)). GIFs keep those four rows and step through the five forecast horizons, with the five input frames held on the left and the current future frame on the right.
GIF paths: figures/gifs/figure3_cm_rank1.gif, figures/gifs/figure3_cm_rank2.gif, figures/gifs/figure3_cr_rank1.gif, figures/gifs/figure3_cr_rank2.gif.

Figure 3: Seed-0 Model A. Mirror ranks are test videos 9347 ($C_M = 0.00624$) and 9257 ($C_M = 0.00466$). Rotation ranks are 9118 ($C_R = 0.00729$) and 9493 ($C_R = 0.00676$). Each panel has four rows: $X \mid f(X)$; $T(X) \mid T(Y)$; $T(X) \mid f(T(X))$; $T(X) \mid T(f(X))$. GIFs: `figures/gifs/figure3_cm_rank1.gif`, `figures/gifs/figure3_cm_rank2.gif`, `figures/gifs/figure3_cr_rank1.gif`, `figures/gifs/figure3_cr_rank2.gif`. Rows 3 and 4 disagree, and the rotation panels disagree more.

Figure 4. Per-video consistency on the 1,000 test videos

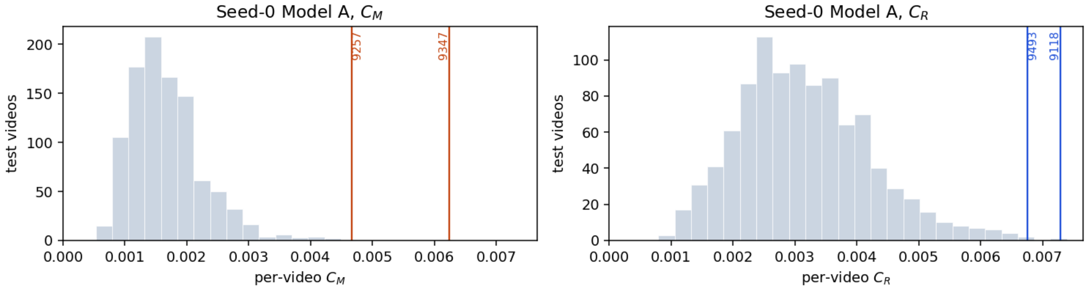


Figure 4. Histograms of per-video C_M and C_R for seed-0 Model A on the 1,000 test videos, dataset indices 9000-9999. Vertical lines mark the Figure 3 videos. C_M lines: 9347 (0.00624), 9257 (0.00466). C_R lines: 9118 (0.00729), 9493 (0.00676).

Figure 4: Per-video C_M and C_R of seed-0 Model A on the 1,000 test videos. The Figure 3 videos are marked. Rotation gaps are larger on average.

Table 4: Speed comparison at $t = 1$. $E(5)$ at 3× is omitted. Blank is deterministic.

	Model A	Blank	Copy-last
$E(2)$, 1×	0.0186 (0.0191, 0.0181)	0.0426	0.0548
$E(2)$, 2×	0.0448 (0.0449, 0.0448)	0.0425	0.0661
$E(2)$, 3×	0.0520 (0.0528, 0.0511)	0.0426	0.0703
$E(5)$, 1×	0.0221 (0.0225, 0.0216)	0.0426	0.0641
$E(5)$, 2×	0.0490 (0.0486, 0.0493)	0.0425	0.0719

Figure 5. Seed-0 Model A, test video 9883, speeds 1x 2x 3x

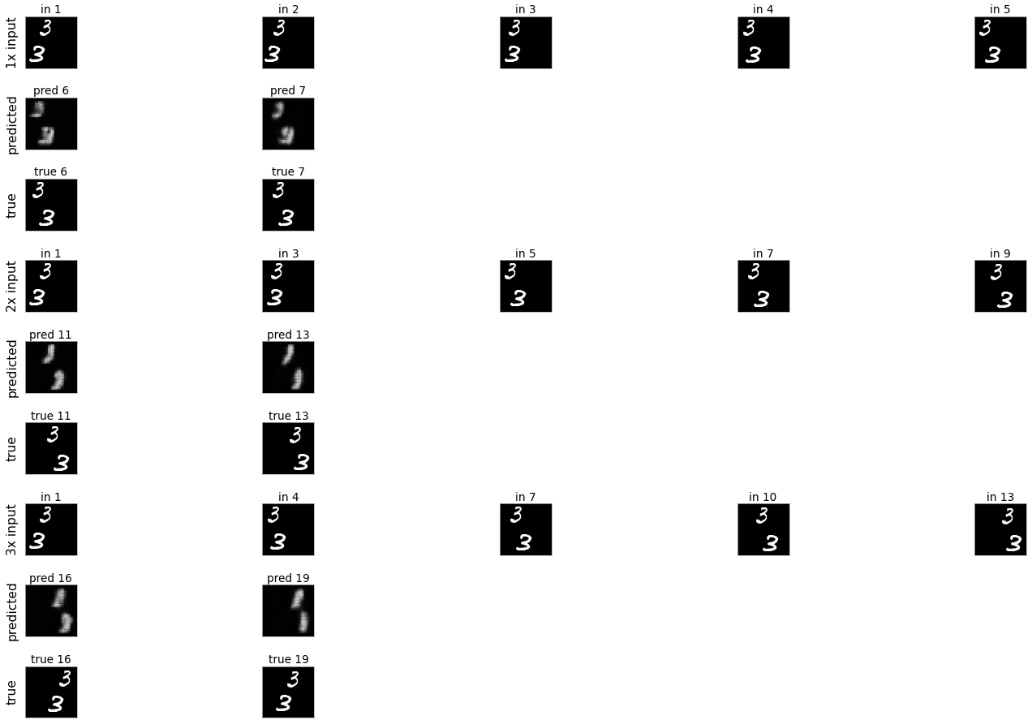


Figure 5. Seed-0 Model A on test video 9883. Each speed shows five inputs and the first two predicted and true futures at t=1. Non-overlap: every displayed true frame has exactly two 8-connected components of pixels above 0.1, each of area at least 80 (area>=80 and combined frame-5-to-7 displacement>=4). 35 videos qualified. Minimum centroid distance 31.19 px.

Figure 5: Seed-0 Model A on test video 9883. Five inputs and the first two predicted and true futures at 1×, 2×, and 3×. The displayed true frames keep the two digits apart. At 2× and 3× the forecast is one blob.

Q2.6. Blank and Copy-last have $C_M = C_R = 0$, and their transformed errors equal their $E(5)$, 0.0426 and 0.0641. A forecaster that ignores its input is also perfectly consistent, so a low C_T on its own can mean a useless forecast. C_T is reported next to E_T.

Q2.7. On video 9347, seed 0, row 3 still has two digits at horizon 5, with the second digit too far down and not far enough left, while row 4 is one blob. The per-video C_M is 0.00624.

Q2.8. Rotation is harder than mirroring for Model A: $C_R = 0.00308$ against $C_M = 0.00183$, and $E_R = 0.0238$ against $E_M = 0.0227$. The speed is the same under both, and every direction bin is already occupied, so the transformed motion is familiar. A 90° turn lays the digit on its side, a shape the 1× training did not show, while a flip keeps vertical strokes vertical. The larger gap is appearance.

Q2.9. Model A falls apart at 2×, and 3× is worse: $E(2)$ goes from 0.0186 at 1× to 0.0448 at 2×, above Blank's 0.0425, and to 0.0520 at 3×. The typical 2× step is 6.64, above the 1× 99th percentile of 4.67, so almost none of the training motion was that fast, even though 6.64 is inside the 8-pixel reach. The soft blob is the conditional mean over positions the model cannot pin down. In Figure 5 the 1× forecast follows both digits, and at 2× and 3× it is one soft blob.

Q2.10. The baseline does not separate a missing fast example from a step beyond reach, because Model A never saw 2× or 3× and its reach is 8 pixels. Models C and D are that experiment: C keeps A's architecture and adds speeds $\{1, 2, 3\}$, and D keeps that schedule and lengthens the reach. If C fixes 3×, the failure was unfamiliar motion. If C still fails at 3× and D does not, it was reach.

2.6 Q2.11

This answer was committed as `ede80b4` at 16:20 on 27 October 2026, before training B, C, and D. It is unchanged.

Model B should close the mirror and rotation gaps, the rotation gap most of all, because it trains on the group G, and it should not fix the speed failure because it never sees 2× or 3×. Model C should improve 2× if that error is unfamiliar motion inside the 8-pixel reach, and it should leave 3× broken, because a typical 3× step is 9.95 pixels and C still has Model A's reach. Model D, with reach 16 and the same speed schedule as C, is the one that should fix the 3× failure, and the part of the 2× failure that sits near the edge of 8 pixels.

2.7 Models B, C, and D

Model B is Model A trained at 1× with one uniform draw from the eight elements of G applied to the input and the target. Model C is Model A trained with k drawn from $\{1, 2, 3\}$ and no symmetry augmentation. Model D is the same schedule as C, with dilation 2 and padding 2 on both 3×3 layers. Dilation does not add weights, so D also has 480,784 parameters, under the cap of 721,176. No attention is used.

Q2.12. Model D's one-step reach is 16 original pixels, against Model A's 8. A 3×3 kernel with dilation 2 reads offsets $-2, 0, +2$, so one layer moves 2 folded cells, which is 8 original pixels. The new input passes through both layers, and layer 2 reads layer 1's new hidden state, so the longest path is 4 folded cells, $4 \times 4 = 16$. The path is $u_\tau \to$ layer-1 dilated $3 \times 3 \to h_\tau^{(1)} \to$ layer-2 dilated $3 \times 3 \to \hat{x}_{\tau+1}$. The 1×1 head and the shuffles add nothing. The previous state of layer 2 travels only one dilated kernel, which is 8 original pixels. One path moves an even number of folded cells, 0, 2, or 4, so an odd offset is reached only by combining paths. The reach is the longest path. Dilation adds no weights; a 5×5 kernel on both layers would pass the cap of 721,176.

Table 5: Test metrics for A–D. Consistency is at 1×. $E(2)$ at 3× uses the first two horizons only. Training and inference times are seconds on a Tesla T4.

	A	B	C	D
$E(5)$, 1×	0.0221 (0.0225, 0.0216)	0.0223 (0.0226, 0.0221)	0.0267 (0.0268, 0.0267)	0.0287 (0.0283, 0.0290)
C_M	0.00183 (0.00168, 0.00198)	0.00106 (0.00108, 0.00105)	9.30×10^{-4} (8.82×10^{-4}, 9.77×10^{-4})	9.81×10^{-4} (9.90×10^{-4}, 9.71×10^{-4})
E_M	0.0227 (0.0232, 0.0223)	0.0224 (0.0227, 0.0221)	0.0269 (0.0270, 0.0269)	0.0289 (0.0286, 0.0292)
C_R	0.00308 (0.00315, 0.00302)	0.00132 (0.00126, 0.00139)	0.00205 (0.00188, 0.00222)	0.00181 (0.00166, 0.00197)
E_R	0.0238 (0.0242, 0.0234)	0.0223 (0.0226, 0.0219)	0.0279 (0.0279, 0.0278)	0.0296 (0.0292, 0.0299)
$E(2)$, 1×	0.0186 (0.0191, 0.0181)	0.0186 (0.0187, 0.0184)	0.0230 (0.0230, 0.0230)	0.0263 (0.0261, 0.0266)
$E(2)$, 2×	0.0448 (0.0449, 0.0448)	0.0446 (0.0444, 0.0448)	0.0288 (0.0288, 0.0289)	0.0293 (0.0289, 0.0297)
$E(2)$, 3×	0.0520 (0.0528, 0.0511)	0.0506 (0.0496, 0.0515)	0.0335 (0.0336, 0.0334)	0.0316 (0.0311, 0.0321)
Parameters	480,784	480,784	480,784	480,784
s/epoch	11.3	11.6	12.4	19.9
Inference (s)	0.461	0.464	0.453	0.720

Q2.13. The mirror and rotation gaps are appearance: Model B cuts C_M from 0.00183 to 0.00106 and C_R from 0.00308 to 0.00132, with $E(5)$ still 0.0223 and $E(2)$ still 0.0446 at 2× and 0.0506 at 3×. The 2× failure is unfamiliar motion: Model C, at the same 8-pixel reach, has $E(2) = 0.0288$ at 2× against Model A's 0.0448 and Model B's 0.0446, and Model D is level with C at 0.0293, inside the 6.64-pixel typical step. At 3×, Model C moves $E(2)$ from 0.0520 to 0.0335, and Model D reaches 0.0316. Most of that drop is from seeing the fast clips. The further drop from dilation is about 0.0019, close to D's extra error on the original clips. D is also worse than C at 1× (0.0287 against 0.0267) and at 2× (0.0293 against 0.0288). If C had still failed at 3× while D succeeded, the failure would have been reach, because the two models saw the same speeds. That is not the table. More data cannot carry information further than the architecture allows: data only changes the weights, and no weight moves information along a path the network does not have. Within one step the 3×3 stack only sees cells inside its reach.

Q2.14. Model B's $E(5)$ at 1× is 0.0223 against Model A's 0.0221, so the symmetry training did not cost the original clips, and it cut C_M to 0.00106 and C_R to 0.00132. Model C's $E(5)$ rises to 0.0267, a rise of 0.00467, while $E(2)$ falls by 0.0160 at 2× and by 0.0185 at 3×, and $E(2)$ at 1× rises from 0.0186 to 0.0230. Model D's $E(5)$ is 0.0287; against C the 1× cost is about 0.00193 and the 3× gain is about 0.00185, and at 2× D's 0.0293 does not beat C's 0.0288. C's 1× cost of 0.00467 is worth the 2× and 3× gains of 0.0160 and 0.0185. D's extra reach is not: the 3× gain of about 0.00185 matches the 1× cost. B is worth it, because the gaps fall and $E(5)$ stays at 0.0223 against A's 0.0221.

2.8 The final model

Table 5 supports D+B. Model B cut the rotation gap without moving $E(5)$, and Model D's rotation gap (0.00181) was still larger than its mirror gap. D+B is Model D trained with both the speed draw and one shared group element on each window. Its $E(5)$ is 0.0285 (0.0287, 0.0282), slightly under D. Its gaps are $C_M = 7.53 \times 10^{-4}$ and $C_R = 9.89 \times 10^{-4}$, with $E_M = E_R = 0.0285$.

Its $E(2)$ is 0.0261, 0.0290, and 0.0313 at 1×, 2×, and 3×. The speed gains survive. The final model is that network wrapped as

$$\bar{f}(X) = \frac{1}{8} \sum_{g \in G} g^{-1}\big(f(g(X))\big).$$

For any $h \in G$, right multiplication by h permutes G. The group element only rearranges pixels, so it commutes with the average and factors out, and $\bar{f}(h(X)) = h(\bar{f}(X))$. Since M and R are in G, $C_M = C_R = 0$ exactly. The measured gaps are the floating-point residual.

Table 6: Symmetrised forecasters. Each learned entry is the mean, with seed 0 and seed 1 on the next line. Training time is the underlying model's seconds per epoch. Inference includes all eight transforms. C_M and C_R are in scientific notation.

	Sym. B	Sym. D	Sym. D+B (final)
$E(5)$, 1×	0.0218	0.0285	0.0281
	(0.0221, 0.0214)	(0.0282, 0.0289)	(0.0284, 0.0279)
C_M	8.43×10^{-17}	5.47×10^{-17}	5.65×10^{-17}
	$(8.13 \times 10^{-17}, 8.72 \times 10^{-17})$	$(4.77 \times 10^{-17}, 6.17 \times 10^{-17})$	$(5.57 \times 10^{-17}, 5.74 \times 10^{-17})$
E_M	0.0218	0.0285	0.0281
	(0.0221, 0.0214)	(0.0282, 0.0289)	(0.0284, 0.0279)
C_R	6.76×10^{-17}	4.26×10^{-17}	4.48×10^{-17}
	$(6.49 \times 10^{-17}, 7.02 \times 10^{-17})$	$(3.69 \times 10^{-17}, 4.84 \times 10^{-17})$	$(4.39 \times 10^{-17}, 4.56 \times 10^{-17})$
E_R	0.0218	0.0285	0.0281
	(0.0221, 0.0214)	(0.0282, 0.0289)	(0.0284, 0.0279)
$E(2)$, 1×	0.0181	0.0263	0.0259
	(0.0183, 0.0180)	(0.0260, 0.0266)	(0.0261, 0.0257)
$E(2)$, 2×	0.0438	0.0291	0.0286
	(0.0437, 0.0439)	(0.0288, 0.0295)	(0.0288, 0.0285)
$E(2)$, 3×	0.0495	0.0314	0.0309
	(0.0487, 0.0503)	(0.0309, 0.0319)	(0.0311, 0.0306)
Parameters	480,784	480,784	480,784
s/epoch	11.6	19.9	20.7
Inference (s)	3.81	5.24	4.95
	(4.42, 3.20)	(5.20, 5.28)	(4.90, 5.00)

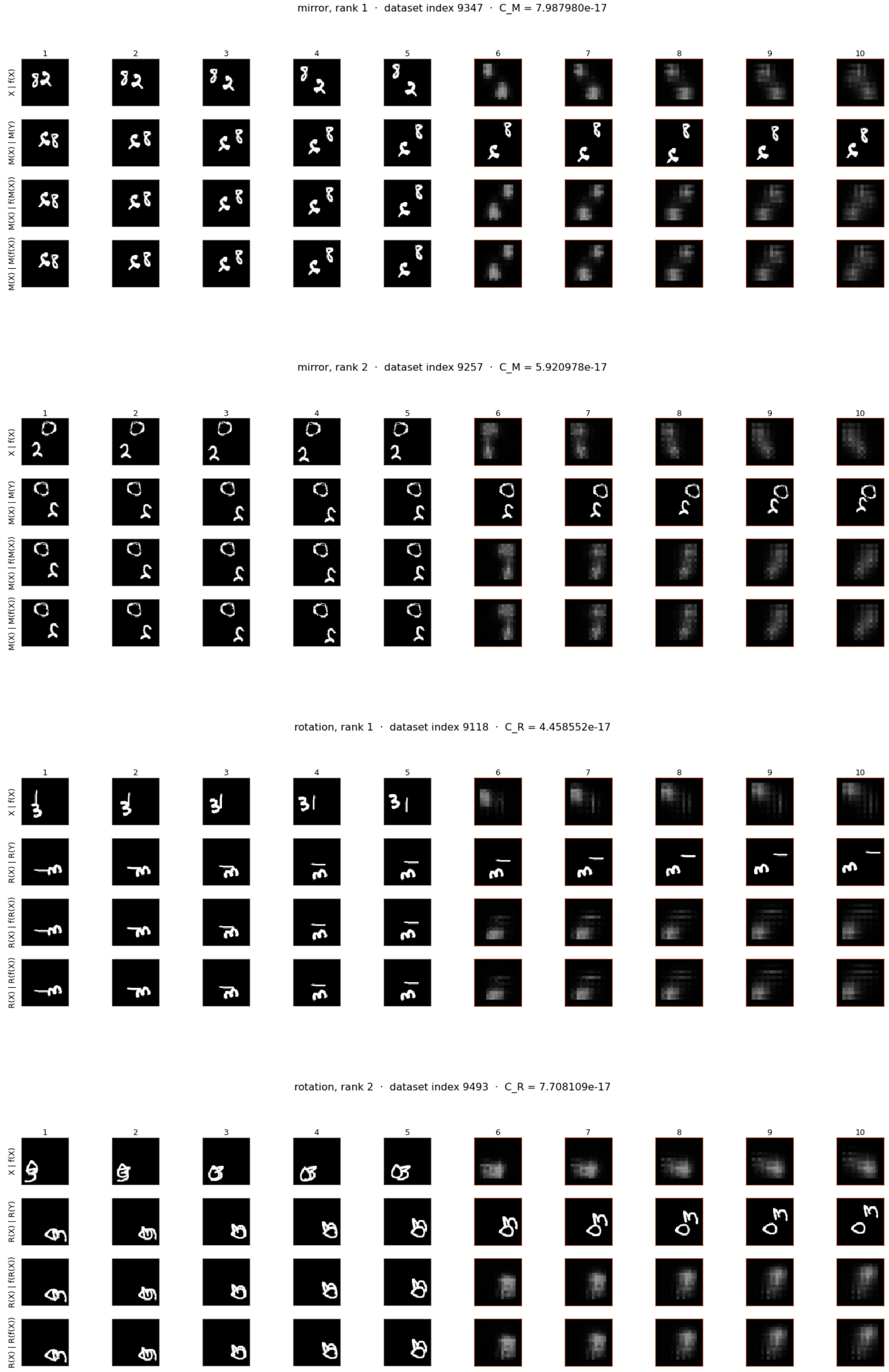


Figure 6: Symmetrised D+B, seed 0. (a) The Figure 3 videos: mirror 9347 and 9257, rotation 9118 and 9493. Rows 3 and 4 agree. GIFs: `figures/gifs/figure6a_cm_rank1.gif`, `figures/gifs/figure6a_cm_rank2.gif`, `figures/gifs/figure6a_cr_rank1.gif`, `figures/gifs/figure6a_cr_rank2.gif`.

Figure 6b. Seed-0 symmetrised D+B, dataset 9883, speeds 1x 2x 3x

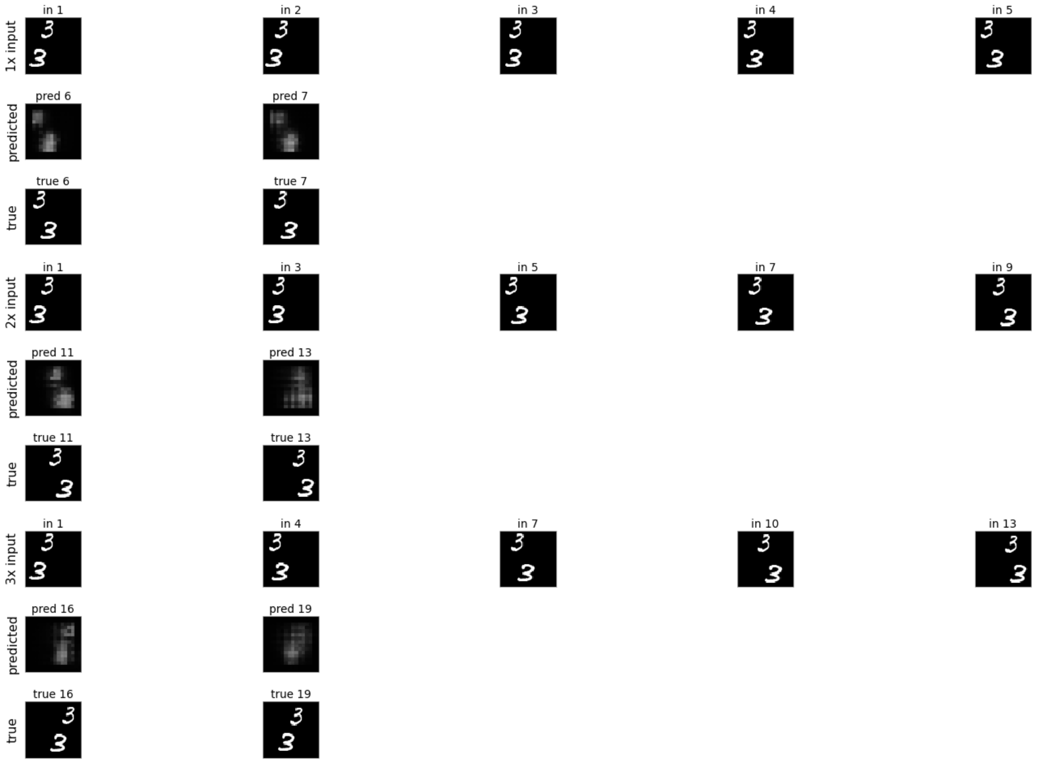

1x input
in 1
in 2
in 3
in 4
in 5
predicted
pred 6
pred 7
true
true 6
true 7
2x input
in 1
in 3
in 5
in 7
in 9
predicted
pred 11
pred 13
true
true 11
true 13
3x input
in 1
in 4
in 7
in 10
in 13
predicted
pred 16
pred 19
true
true 16
true 19

Figure 6b. Seed-0 symmetrised D+B on dataset index 9883, the same video as Figure 5. Each speed shows five inputs and the first two predicted and true futures at t=1. At 3x only those two futures exist inside the clip.

Figure 6: Symmetrised D+B, seed 0. (b) Test video 9883, the same video as Figure 5. The wrapper makes the two mirror orders agree, and the merged digit stays merged.

Q2.15. The wrapper is exact and the learned symmetry is not: Model B still has $C_M = 0.00106$ and $C_R = 0.00132$, and D+B has 7.53×10^{-4} and 9.89×10^{-4}, while the wrapped gaps are of order 10^{-17} and $E_M = E_R = E(5)$. Wrapped B's $E(5)$ is 0.0218 against learned B's 0.0223, and wrapped D+B's is 0.0281 against 0.0285; the mean squared spread of the eight forecasts is 3.91×10^{-4} (worst video 1.13×10^{-3}), so the average did not raise the test error. Built-in symmetry would hurt when the eight forecasts put a digit in different places and the average smears it. The spread here is 3.91×10^{-4}, so that did not happen. Inference is the cost: wrapped B takes 3.81 s (4.42, 3.20) against learned B's 0.464 s (0.517, 0.412), wrapped D takes 5.24 s (5.20, 5.28) against 0.720 s (0.725, 0.715), and wrapped D+B takes 4.95 s (4.90, 5.00) against 0.712 s (0.718, 0.705), about 8.2, 7.3, and 7.0 times as long, with D+B's training time still 20.72 s per epoch (20.67, 20.78).

Q2.16. On video 9347, the video in Q2.7, symmetrised D+B makes rows 3 and 4 the same image: the per-video C_M is 7.99×10^{-17}. Horizon 5 is still wrong. The truth has two digits and the forecast is one blob between them, because the underlying forecast had already merged and the wrapper copies that merge into both orders. The wrapper fixed the disagreement between rows 3 and 4. Nothing fixed the merge.

Q2.17. The final model learned the speeds that were in training, and it did not learn to keep two digits apart. Symmetrised D+B has mean $E(2)$ 0.0259 at 1×, 0.0286 at 2×, and 0.0309 at 3×, against Model A's 0.0186, 0.0448, and 0.0520, while Blank stays near 0.0426. On video 9883 the truth keeps two digits. The prediction is one component by frame 7 at 1× and by frame 11 at 2×, and at 3× it is one blob between the two paths. A skeptic should require both centroids to track the truth on a non-overlapping clip at a speed the training loop never drew. The test it has not passed is that separation on video 9883.

AI use disclosure

I used Cursor (Composer) while coding and while drafting this report.
Coding prompts (summary): set up the repo off Google Drive; run Table 1; check that Model A has 480,784 parameters; train on a Kaggle Tesla T4; make the optical-flow figure; design the river model; design Model D; train B, C, D, and D+B; symmetrise the forecaster; write the report; shorten the report and draw the Question 1 architecture; fill the missing answers flagged by the audit; enlarge Figure 6a and redraw the river diagram with the masks.
Model help on code: wrote `transforms.py`, `windows.py`, `data.py`, `metrics.py`, `convlstm.py`, `model_a.py`, `model_d.py`, and `symmetrise.py`; wrote the training scripts `train_a.py`, `train_b.py`, `train_c.py`, `train_d.py`, and `train_db.py`; ran Table 1, Figure 1, and every training and evaluation job on Kaggle (Tesla T4). Model D's dilation and the choice to train D+B were made in that work.
My edits on code: checked that Table 1 matches, that Model A and Model D each have 480,784 parameters, that Q2.5 and Q2.11 were pushed before the experiments they precede, and that every reported test number matches the result JSON files.
Question 1: Cursor drafted the six headings. The history length and the six-step horizon are assumptions A1 and A2, because the brief leaves both unspecified. I checked that the section separates those assumptions from the given river, and that future rain is not an input while an already announced release is. The diagram is the TikZ figure, and the six headings were shortened.
Question 2: Cursor implemented the model, training, and figure scripts and ran them on Kaggle (Tesla T4). I checked Table 1, the 480,784 parameter count, and that the reported test numbers come from the result files. Q2.5 was committed as `10cb5f4` at 14:35 on 27 October 2026, before the mirror, rotation, and speed tests. Q2.11 was committed as `ede80b4` at 16:20 the same day, before training B, C, and D.